

PROFILE SUMMARY

- Computer Science graduate with strong knowledge of **Python, SQL, Machine Learning, LangChain, React.js, AWS, and ETL**, with experience developing AI applications and data-driven solutions.
- Skilled in **data cleaning, preprocessing, ETL, feature engineering, model training, hyperparameter tuning, and model evaluation**.
- Proficient in **Pandas, NumPy, TensorFlow, Power BI, Microsoft Excel, and React.js** for data analysis, visualization, and application development.
- Strong understanding of **DSA, OOP, DBMS, SQL, data modeling, relational databases, and data validation**.
- Hands-on experience building **NLP pipelines and LangChain-based AI applications** for intelligent and scalable solutions.
- Experienced in **workflow automation** using Microsoft Power Automate, Power Apps, and Zapier.
- Familiar with **Microsoft Azure services**, including Azure Virtual Machines, Azure Storage, and Azure SQL Database.
- Interested in **AI/ML, Generative AI, data engineering, full-stack development, cloud technologies, and scalable system design**.

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, C
- **Web Technologies:** HTML, CSS, React.js
- **Python Libraries & Frameworks:** Pandas, NumPy, Scikit-learn, TensorFlow, LangChain
- **Machine Learning & NLP:** Supervised Learning, Natural Language Processing (NLP), Feature Engineering, Data Cleaning & Preprocessing, Model Training, Hyperparameter Tuning, Model Evaluation
- **Databases:** MySQL, Oracle Database, SQL
- **Data Engineering:** ETL (Extract, Transform, Load), Data Cleaning, Data Preprocessing, Data Validation, Data Modeling
- **Cloud & Platforms:** AWS S3, AWS EC2, Azure Virtual Machines, Azure Storage, Azure SQL Database
- **AI & Generative AI:** LangChain, LLM Applications, RAG, Prompt Engineering, NLP Pipelines
- **Automation & Low-Code:** Microsoft Power Automate, Power Apps, Zapier
- **Data Visualization & Analytics:** Microsoft Power BI, Tableau, Microsoft Excel
- **Development & Tools:** Git, FastAPI, REST APIs
- **Core CS Concepts:** Data Structures & Algorithms, Object-Oriented Programming (OOP), Database Management Systems (DBMS)

- **Employee Leave Management System**

Description: An employee leave management application designed to streamline leave requests, approval workflows, status tracking, and notifications using Microsoft Power Platform.

Tech Stack: Power Apps Canvas Apps, Power Automate, Dataverse, SharePoint, AI-assisted Validation

Responsibilities:

- Developed a responsive Power Apps Canvas App for employees to submit and track leave requests.
- Designed user interfaces for Employees, Managers, and HR with role-based access.
- Built multi-level approval workflows using Power Automate for manager and HR approvals.
- Integrated Dataverse and SharePoint for storing and managing employee and leave-related data.
- Implemented automated email notifications and reminders for leave submissions, approvals, and pending requests.
- Added AI-assisted validation to identify incomplete or missing information before approval.
- Performed testing, debugging, access validation, data integration, and technical documentation.

- **AI Content Studio – Social Media & Website Support**

Description: An AI-powered content creation and editing application that helps generate, rewrite, and improve social media captions, website content, and marketing copy. It supports platform-ready content for Instagram, Facebook, LinkedIn, X, and websites.

Tech Stack: React JS, Tailwind CSS, Node.js, MongoDB, OpenAI API, LangChain

Responsibilities:

- Developed a responsive and user-friendly frontend using React JS and Tailwind CSS.
- Built backend functionality using Node.js for AI content generation.
- Integrated OpenAI API and LangChain for prompt-based content creation.
- Created features to generate social media captions, website copy, and marketing content.
- Added a live editor for reviewing and editing generated content.
- Stored generated content history using MongoDB.
- Added options to reuse, delete, and copy generated content.
- Improved UI/UX with a clean layout and simple workflow.

- **AI Learning Management System**

Description: Developed an AI-powered Learning Management System with Student, Teacher, and Admin role-based access. The platform enables AI-assisted course creation, document-based learning, intelligent question answering, progress tracking, and personalized learning experiences through Retrieval-Augmented Generation (RAG).

Tech Stack: Python, FastAPI, React.js, LangChain, OpenAI API, MongoDB, ChromaDB, JWT, Tailwind CSS

Responsibilities:

- Designed and developed an AI-powered Learning Management System with secure authentication and role-based access for Students, Teachers, and Admins.
- Built AI-assisted course creation, allowing teachers to generate course descriptions, chapters, quizzes, and learning materials.
- Implemented a RAG pipeline using LangChain and ChromaDB for document-based question answering.
- Developed REST APIs using FastAPI to manage users, courses, documents, enrollments, and AI interactions.
- Built a responsive and user-friendly frontend using React.js and Tailwind CSS.
- Integrated MongoDB for managing users, courses, progress tracking, and learning resources.
- Implemented role-specific features such as course management for teachers, enrollment and progress tracking for students, and user management for admins.
- Improved learning experience by providing AI-generated summaries, explanations, and personalized study assistance.

- **Social Media-Based Suicidal Ideation Detection**

Description: Developed an NLP-based text classification system to analyze large-scale social media datasets and identify patterns related to suicidal ideation.

Tech Stack: Python, Scikit-learn, TensorFlow, BERT, Transformer Based Model, Pandas

Responsibilities:

- Designed and implemented an end-to-end NLP pipeline for data preprocessing and model training.
- Performed text cleaning, tokenization, lemmatization, and vectorization using TF-IDF and transformer embeddings.
- Implemented and fine-tuned transformer-based architectures for contextual text understanding.
- Conducted hyperparameter tuning and comparative model evaluation.
- Built structured modular workflow for training, validation, and testing.

- **Atlas – Research & Knowledge Agent**

Description: Developed an AI-powered research assistant that enables users to upload documents, manage knowledge sources, and generate context-aware responses using Retrieval-Augmented Generation (RAG).

Tech Stack: FastAPI, Python, MongoDB, OpenAI API, LangChain, LangGraph, RAG, React.js, TypeScript

Responsibilities:

- Developed secure backend APIs and JWT-based authentication using FastAPI.
- Implemented document ingestion, embedding, and semantic search capabilities.
- Built RAG workflows for intelligent knowledge retrieval and question answering.
- Developed responsive frontend interfaces for workspace and AI chat management.
- Integrated OpenAI models for research assistance and content generation.
- Deployed and maintained the application on Render and Vercel.

• Blinkit Sales Dashboard

Description: Developed a sales dashboard to track and visualize sales performance, helping users understand key business metrics and trends.

Tech Stack: Tableau, SQL, Python

Responsibilities:

- Collected and cleaned sales data for analysis and reporting.
- Created interactive dashboards to visualize sales trends and order performance.
- Used SQL queries to extract and organize data from datasets. SQL queries to extract and organize data from datasets.
- Added filters and charts for better user interaction and analysis.
- Analyzed sales data to identify patterns and support business decisions.

INTERNSHIP EXPERIENCE

Summer Intern | Jun 2025 – Jul 2025

Jadavpur University, Kolkata

Description: Worked on building sentiment analysis models to classify textual data into sentiment categories using supervised Machine Learning techniques.

Tech Stack: Python, Scikit-learn, Pandas, NumPy, NLP Techniques

Responsibilities:

- Collected and processed real-world textual datasets for sentiment classification.
- Performed text preprocessing including cleaning, tokenization, stopwords removal, and vectorization (TF-IDF, word2vec).
- Implemented and compared multiple supervised learning algorithms for sentiment prediction.
- Applied feature engineering and hyperparameter tuning techniques.
- Followed standard Machine Learning experimentation workflow including training, validation, and evaluation.

EDUCATION

• B.Tech in Computer Science & Engineering

MCKV Institute of Engineering, MAKAUT, West Bengal

CGPA: **9.31** / 10

Year: 2022 – 2026